

This roadmap outlines the development of the Banking Management System using an Agile methodology. We will operate in **two-week sprints** over a total project duration of **20 weeks (5 months)**.

Project Duration: 20 Weeks

Milestones

1. **M1: Discovery & Architecture (Weeks 1-2)** – Infrastructure setup and security compliance design.
2. **M2: Foundation Phase (Weeks 3-6)** – Core banking engine, Auth, and Account Management.
3. **M3: Operational Phase (Weeks 7-12)** – Transactions, Loan Workflows, and Admin Controls.
4. **M4: Intelligent Features (Weeks 13-16)** – Fraud Detection and AI Financial Insights.
5. **M5: QA, Compliance & Release (Weeks 17-20)** – Final testing, security audit, and deployment.

Weekly Timeline

Phase	Weeks	Focus Area			
	1-2	3-6	7-12	13-16	17-20
P1	1-2	System Architecture, Database Schema, Security Protocol design (MFA/PCI-DSS compliance).			
P2	3-4	User Auth (MFA), Account creation/retrieval, Balance modules.			
P3	5-6	Transaction Engine (Internal/External transfers), Bill Payment integration.			
P4	7-8	Loan Management workflow (Application, Review, Approval logic).			
P5	9-10	Admin Dashboard (User oversight, logs, audit trails).			
P6	11-12	Performance optimization, Localization, and responsive UI/UX polish.			
P7	13-14	Fraud Detection engine development (Rule-based + threshold triggers).			
P8	15-16	AI Financial Insights engine integration (Spending analysis).			
P9	17-18	Final QA, Security Pen-Testing, Load Testing.			

| **P10** | 19-20 | UAT (User Acceptance Testing), Regulatory Audit, Production Release. |

Deliverables

- **System Architecture Design Document:** Security layers and compliance roadmap.
- **Production-Ready Backend API:** Secure REST/GraphQL endpoints with MFA.
- **Client Interfaces:** Web portal for Customers, Staff, and Admins.
- **Loan Approval Workflow:** Automated lifecycle from submission to payout.
- **Fraud Detection Engine:** Real-time monitoring with automated alerts.
- **AI Budgeting Module:** Daily transactional analysis tool.
- **Documentation:** User manuals, API docs, and Audit/Compliance reports.

Risk Management

| Risk | Impact | Mitigation |

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| **Data Breach** | Critical | End-to-end encryption, automated rotating keys, and monthly third-party pen-testing. |

| **False Fraud Positives** | High | Implement a "Review & Approve" queue for customers; fine-tune thresholds after launch. |

| **Compliance Bottleneck** | Medium | Engage legal/compliance consultants during Phase 1 to ensure audit logs meet regulatory standards. |

| **AI Inaccuracy** | Medium | "Human-in-the-loop" requirement for high-impact financial recommendations. |

| **Integration Failure** | Medium | Use mock APIs for the third-party KYC provider during early sprints to decouple progress. |

Agile Management Notes:

- **Scrum Ceremonies:** Daily Stand-ups, Bi-weekly Sprint Planning, Sprint Review (Demo), and Retrospectives.
- **Quality Assurance:** Automated Unit testing, Integration testing for transactions, and manual UAT for loan workflows.
- **Deployment:** CI/CD pipeline (Jenkins/GitHub Actions) with automated regression testing.
- **Communication:** Weekly stakeholder syncs to address the "Open Questions" (e.g., Open Banking integration scope).